

Substrate SM Program — honest-state ledger (claim-by-claim derived/assigned/matched scorecard) v0.1

Keeper

2026-05-28 Thursday EDT

Honest-state ledger v0.1

Status legend: DERIVED (mechanism proved) · IDENTIFIED (matches <1%, mechanism partial) · MATCHED (value coincides, mechanism not forced) · ASSIGNED (hand-labeled; not yet derived) · FRAMEWORK (structure stated, not closed) · RETRACTED · UNBUILT.

L1 — The algebra (engine)

Claim	Tier	Status	Rests on	Flips when
Substrate Hall algebra = $U_q^+(B_2)$ at $q=2$; Serre constants = primaries $\{N_c, n_c, g, N_c \cdot g\}$	D	DERIVED / RIGOROUS	E0 (Toy 3597) + A1 §3, integrality	— (stands; the strongest result in the engine)
“Substrate IS this algebra” (B_2 from D_{IV}^5 type-IV)	I	IDENTIFIED	A1 geometry $\rightarrow B_2$	the canonical basis + dictionary ground it
4/4 Hopf pieces (mult+coproduct, canonical basis, R-matrix, negative part) = Drinfeld double	D	RIGOROUS AS ALGEBRA	standard quantum-group theory (Lyra)	— (math stands; $\neq$ physics derived)
“Goal 1 structurally complete”	—	WARRANTED, with caveat	the 4 pieces exist	caveat: “structure complete $\neq$ physics derived” must travel with it

L2 — Kinematics (what particles are)

Claim	Tier	Status	Flips when
5-tuple particle taxonomy	partial D / partial ASSIGNED	LEPTON SECTOR DERIVED (Lyra #416 v0.1, 18 entries — every static axis derived: σ_{BF} /region/chirality/charge/particle- anti; generation indexed but per #414); photon/Higgs single-particle clean; quark + gauge sectors still ASSIGNED pending the bulk-color mechanism (see Gates) DERIVED (solid)	quark/gauge flip together with bulk-color resolution —
3 colors = $h^1(B_2) = N_c = 3$	D		
3 generations = $h-1 = 3$	—	MATCHED (the gate)	see “Gates” below
charge = S^1 winding / SO(2) weight; fractional charge forced by Z_3 closure	D	DERIVED (mechanism)	— (per-particle assignment is ASSIGNED)
chirality / parity violation = D_{IV}^5 Hermitian $\rightarrow$ complex structure $J =$ SO(2) factor of $K \rightarrow$ holomorphic Hardy space picks one sign of J $\rightarrow SU(2)_L =$ J-compatible half of Shilov stabilizer $SO(4) \rightarrow$ left-handed weak coupling, right-handed weak singlets	I	FRAMEWORK-PLUS — Lyra L3 geometric mechanism; the substrate’s <i>complex orientation</i> IS the parity choice (“Nature doesn’t waste the i” — one SO(2) does four jobs: Hermitian / U(1) charge / Shilov direction / parity)	per-particle L/R labeling pending SO(5,2) $\rightarrow$ Lorentz embedding pin (Lyra honestly flagged)
confinement (no Shilov value for non-singlets)	I	DERIVED, FRAMEWORK-PLUS	coupling map multi-week

L3 — Dynamics (the process engine)

Claim	Tier	Status	Flips when
fusion = Hall product (extension-counting, GF(2))	D	DERIVED AS ALGEBRA (E2/3600)	physics ID (which product = which reaction) is the dictionary bet
decay = Green coproduct; β -decay conserves Q/B/L via grading	D	DERIVED AS ALGEBRA (E3/3601)	the identification “this coproduct = real β -decay” rides on the dictionary
conservation = grading	D	DERIVED AS ALGEBRA	full-charge-set check pending (E4/L2)
scattering = R-matrix	—	FRAMEWORK (structure exists; amplitudes not extracted)	Lyra L6 extracts amplitudes

Claim	Tier	Status	Flips when
absolute mass scale, Higgs/EWSB, RGE, cross-sections	—	UNBUILT	Lyra L4/L5 + Elie numerics
“Goal 2 mechanism built”	—	WARRANTED for the mechanism	the <i>mechanism</i> is real; the <i>physical model</i> needs the dictionary

L4 — Composites / nuclear

Claim	Tier	Status	Flips when
proton = bulk $k=6$ ($C_2=6$) = YM mass gap; $m_p = 6\pi^5 \cdot m_e$	D	DERIVED (0.01%)	— (anchor)
nuclear shells = SO(5) K-type closures (SO(5) = compact factor of D_{IV}^5)	I	STRUCTURAL / HYPOTHESIS	the SO(5) shell-closure computation (E8): does branching give 2,8,20,28,50,82,126?
magic-number forms (rank·products); 126 = Universal-Q; spin-orbit = C_2/n_C	S	SUGGESTIVE (Grace-flagged)	E8 derivation; small-target/coincidence- denominator caveat until then

L5 — Engineering

SP-30 falsifier designs: EXIST. Control theory: UNBUILT. (Same open dynamics layer as L3.)

L6 — Periodic Table

Claim	Tier	Status	Flips when
v0.2 structure (3 rows $\times$ 4 families + boson block)	—	BUILT	—

Claim	Tier	Status	Flips when
Fundamental-sector selection (the 4-backbone)	FRAMEWORK-PLUS	Grace #413 found 4 of 66 cells Casimir-anchor; Lyra's dictionary (#408, first output) selects the SAME four — reframed to REP-ROLE: they are exactly {trivial + ω_1 vector + ω_2 spinor + adjoint} of $B_2=so(5)$, the canonical building-block set (rank=2 $\rightarrow$ 2 fundamentals + trivial + adjoint = 4). Rep-theory part RIGOROUS (fundamental weights / adjoint of B_2 ; Casimirs 0, 5/2, 4, 6). $so(5)$ -SPECIFIC strengthener: matter = the 4-dim spinor ω_2 = the Dirac 4-spinor (not a generic Lie fact); $\sigma_{BF} \rightarrow$ spinor = unique fermion, trivial/vector/adjoint = bosons (matter/force split).	per-sector PHYSICAL identification (reps = Higgs/photon/lepton/gauge) is the dictionary bet. Lyra's honest reframe: SELECT by rep-role (no tunable freedom); Casimir-anchoring is a CORRELATED consistency check (small reps $\rightarrow$ small Casimirs), NOT independent confirmation. Promoted speculative $\rightarrow$ rep-grounded; DERIVED when the dictionary closes the identifications.
per-particle cell assignments	—	ASSIGNED	the dictionary (#408) flips them to DERIVED
gaps = predictions (no 4th row, no GUT/SUSY/steriles)	—	MATCHED $\rightarrow$ prediction	the row-count forcing closes the no-4th-row gap

The gates (open work)

Gate	Status	Detail
A_sub $\leftrightarrow$ Hall dictionary (THE bottleneck)	UNBUILT, in progress (Lyra L1/#408)	flips L2/L3/L6 from ASSIGNED $\rightarrow$ DERIVED; everything physical rides here

Gate	Status	Detail
Generation-forcing	TENSION RELIEVED, NOT CLOSED (Lyra #412+#414 reframe)	The “4 routes $\rightarrow$ 3” was over-counting; really TWO independent B_2 3’s: (I) Coxeter $h-1 = \Phi^+ -1 = 3$ — UNDERCUT (its δ-tube realization leans ≤ 2). (II) dual-Coxeter $h^\vee = N_c = 3$ — a FORCED B_2 invariant. Lyra re-leaned (I)$\rightarrow$(II), reversing her own prior: color-blindness rules out generations <i>carrying</i> color, not the count <i>equaling</i> N_c via a colorless projection (= Grace’s Track P). So the count-NUMBER 3 is over-determined/forced ($h^\vee=N_c=3$); the IDENTIFICATION “$h^\vee$ counts generations” is the OPEN mechanism (Track P), carrying a real burden: produce TWO <i>independent</i> 3-fold structures (colors AND generations) from ONE invariant. Keeper hold: tension relieved (favored route no longer leans toward 2), but NOT closed — count-number forced, generation-mechanism open. This SUPERSEDES the earlier “two independent 3’s ($h^\vee + h-1$)” framing $\rightarrow$ now “one 3 ($h^\vee$) double-duty.” Tube count #409 now matters less (route I no longer favored).
Generation-forcing — mechanism	MATCHED, leans Coxeter/ δ	gen in affine δ-direction vs N_c-color; degenerate at B_2 ($h-1=h^\vee=3$); Lyra L3 (#412) to turn lean$\rightarrow$argument
α -placement	OPEN	coupling/evaluation at affine level (A1 §5.1)

Gate	Status	Detail
Bulk-color mechanism — v0.2 candidate (Saturday)	CANDIDATE / FRAMEWORK — two-structures burden structurally addressed; SU(3)≠SO(3) burden NEW	Lyra v0.2 Direction (A) sharpened: COLOR = SO(3) sub-vector of the SO(5) bulk holomorphic tangent via rigorous standard branching $5(\text{SO}(5)) \downarrow \text{SO}(3) \times \text{SO}(2) = (3,0) + (1,\pm 1)$. 3-fold is GEOMETRIC (substructure of the bulk K-type's SO(5)-vector content, accessed by further branching — <i>refinement of the v0.1 “must come from bulk” framing</i> : SO(3) IS in K via $\text{SO}(3) \subset \text{SO}(5) \subset \text{K}$, just not surfaced by first-level K-type labels). Two-structures burden ADDRESSED at structural level: color (SO(3) geometric) ≠ generations ($\text{h}^\vee/\text{spinor-tower algebraic}$) — independent mathematical mechanisms (both $\text{B}_2/\text{SO}(5)$ -rooted, but distinct structures). Directions (B) reassigned to generations ($\text{E7 spinor}^3 - 3 \neq \text{manifestly } \text{h}^\vee$); (C) abandoned (no clean Z/3 in D_{IV}^5). NEW LOAD-BEARING BURDEN — the load-bearing open work: $\text{SO}(3) \neq \text{SU}(3)$ (3-dim rotation vs 8-dim gauge). Lyra's “counting-not-symmetry” hypothesis (Track P) says SU(3) phenomenology must EMERGE from 3-direction counting — must derive confinement, asymptotic freedom, 8 gluons, running coupling from substrate SO(3). Multi-week open. Until emergence demonstrated: CANDIDATE only. Joint gate with #414's burden remains valid.
baryogenesis α^4	OPEN (located, not sized)	E↔F asymmetry; Lyra L7 + #399

Standing predictions (falsifiable)

- ν mass ordering = NORMAL (conditional on shared-W_n) — JUNO/DUNE.
- Dirac neutrinos, $m_{\beta\beta} = 0$, no $0\nu\beta\beta$ at any scale — LEGEND-1000/nEXO. (Resolved this session vs the Majorana contradiction.)
- No 4th generation, no GUT, no SUSY, no steriles (table closure / Five-Absence).
- $\hat{\text{B}}_2$ tube count — if Elie's E2 returns $\neq 3$, that's an informative break (tubes ≠ generations → cyclotomic carries it).

Consistency fixes (committed ee35f15)

- gravity = S^1 /EM boundary condition, not a force.
- neutrino character = Dirac (Paper 108 corrected).

Methodology guard (this session)

- Source-Verification Tier calibration (candidate, pending Cal): RECALLED vs VERIFIED-CITED; load-bearing claims capped at MATCHED on a recalled literature value. Motivated by the genus + tube-count sourcing failures (2 in 2 days).

The honest one-line verdict

The engine is real and rigorous as algebra; the dynamics mechanism is built; the proton/charge/confinement/mixing-spine results are genuinely DERIVED — but the bulk of the particle spectrum is still ASSIGNED pending the one dictionary, the generation gate is open on both count and mechanism, and the absolute-scale/Higgs/amplitude/nuclear-derivation work is UNBUILT. We are one dictionary from “physical model,” one computation (E2) from the generation count, and a known list of UNBUILT items from “complete.” That is a strong, honest place — not a finished one.

— Keeper, 2026-05-28 v0.1. Update as the dictionary lands (mass ASSIGNED→DERIVED), as E2 resolves the tube count, and as the $SO(5)$ shell-closure decides the nuclear tier.